

BudgetBuddy

1. Title

BudgetBuddy: A Full-Stack Personal Budget Planning and Expense Management Platform

A personal finance management platform that helps students manage pocket money, track expenses, set savings goals, monitor spending habits, and generate financial reports for better financial discipline.

2. Objective

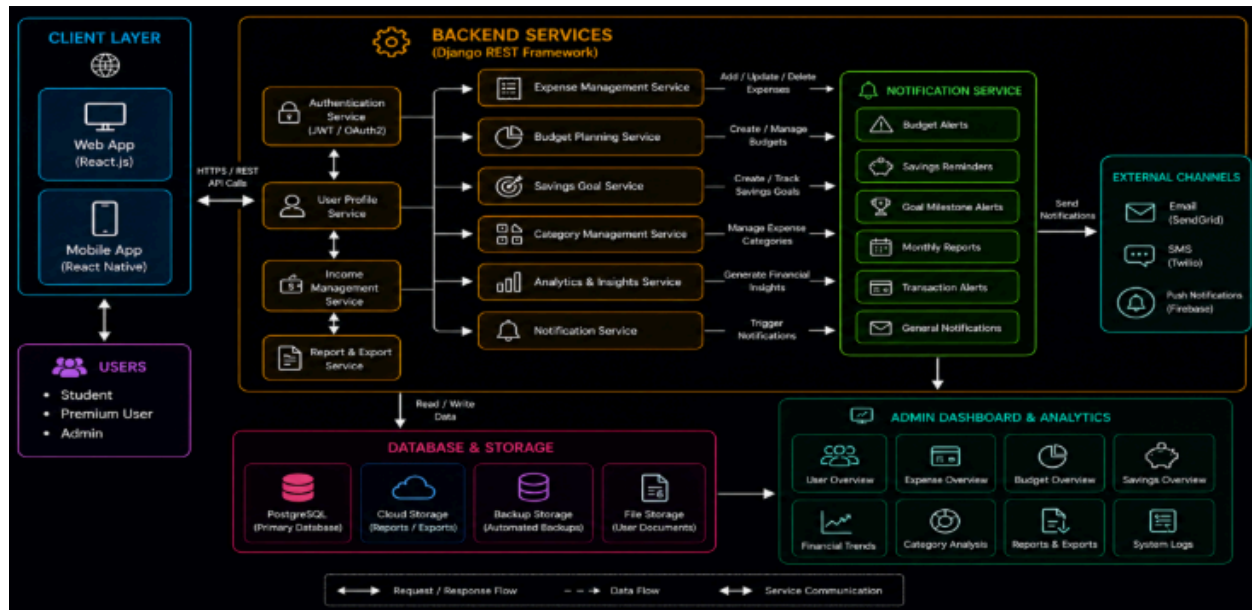
Build a full-stack web application (React.js frontend + Django REST/FastAPI backend) that enables students to track daily expenses, manage monthly budgets, set savings goals, categorize spending, and visualize financial habits through dashboards and analytics.

The platform helps users develop financial awareness and improve money management skills.

Outcomes

- Developed and deployed a full-stack React + Django REST/FastAPI application.
 - Implemented secure user authentication (JWT) and optional OAuth2 login.
 - Built expense and income tracking functionality.
 - Implemented budget planning and savings goal management.
 - Added spending categorization and analytics.
 - Created dashboards for financial insights and expense reports.
 - Implemented notifications for budget limits and savings reminders.
 - Deployed frontend and backend using PostgreSQL in production.
 - Added reporting and data export functionality.
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3. Architecture Diagram



4. Modules to be Implemented

1. User Authentication & Role-Based Access

- JWT Authentication
- OAuth2 Login

Role Management

- Student
- Premium User
- Admin

2. User Profile Management

- Profile creation
- Monthly income setup
- Financial preferences

- Account settings
- Profile management

3. Expense Tracking System

- Add expenses
- Edit expenses
- Delete expenses
- Expense categorization
- Daily transaction history

Categories

- Food
- Travel
- Shopping
- Education
- Entertainment
- Miscellaneous

4. Income Management

- Pocket money tracking
- Scholarship tracking
- Freelance income tracking
- Monthly income reports

5. Budget Planning System

- Monthly budget creation
- Category-wise budget allocation
- Budget utilization tracking

- Budget alerts
- Overspending notifications

6. Savings Goal Management

- Create savings goals
- Goal tracking
- Target amount management
- Progress visualization
- Goal completion tracking

Example Goals

- New Laptop Fund
- Japan Trip Fund
- Emergency Savings

7. Notification & Reminder System

- Budget limit alerts
- Savings reminders
- Monthly report notifications
- Goal milestone alerts

8. Analytics Dashboard

- Expense analytics
- Income analytics
- Savings progress
- Category-wise spending
- Monthly trends
- Financial summaries

9. Reports & Export System

- Generate monthly reports
- Export PDF reports
- Export Excel reports
- Financial summaries
- Expense history reports

10. Final Integration, Testing & Deployment

- System integration
 - Testing
 - Validation
 - Deployment
 - Documentation
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5. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Requirements, Database Design & Backend Setup

Tasks

- Define project scope and user roles.
- Design database schema:
 - Users
 - Profiles
 - Incomes
 - Expenses
 - Budgets

- Savings Goals
- Notifications
- Reports
- Initialize Django REST/FastAPI backend.
- Configure PostgreSQL / SQLite.
- Implement JWT authentication APIs.
- Create React frontend skeleton.

Outcomes

- Backend and frontend architecture setup completed.
- Authentication flow functional.
- Database schema finalized.

Milestone 2: Week 3 & 4 — Expense & Income Management

Tasks

- Implement expense tracking APIs.
- Build income management functionality.
- Create transaction dashboard.
- Add expense categorization.
- Build budget creation system.

Outcomes

- Expense tracking operational.
- Income management functional.
- Budget planning implemented.

Milestone 3: Week 5 & 6 — Savings Goals & Notifications

Tasks

- Implement savings goal management.
- Build notification system.
- Create analytics APIs.
- Add budget alerts.
- Implement report generation.

Outcomes

- Savings goal tracking operational.
- Notification system functional.
- Financial analytics completed.

Milestone 4: Week 7 & 8 — Analytics, Testing & Deployment

Tasks

- Build analytics dashboard.
- Add charts and visualizations.
- Implement testing and validations.
- Deploy frontend/backend using Render, Vercel, or AWS.
- Configure environment variables, SSL, and CORS.

Outcomes

- Fully deployed production-ready application.
- Analytics and reporting systems operational.
- Complete end-to-end financial management workflow demonstrable.

6. Evaluation Criteria

Milestone 1 (Week 2)

- Backend scaffolding complete.

- JWT authentication implemented.
- Database schema finalized.
- Frontend authentication flow working.

Milestone 2 (Week 4)

- Expense tracking operational.
- Income management functioning.
- Budget planning system implemented.

Milestone 3 (Week 6)

- Savings goal management operational.
- Notification system functional.
- Analytics workflow completed.

Milestone 4 (Week 8)

- Fully deployed frontend and backend.
 - Dashboard and reporting systems operational.
 - Demonstrable end-to-end financial management workflow.
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7. Tools & Tech Stack

Programming Language

Backend

- Python
- Django REST Framework / FastAPI

Frontend

- JavaScript
- React.js

Database

Local Development

- SQLite

Production

- PostgreSQL

Authentication

- JWT (SimpleJWT)
- OAuth2 (Google/GitHub)

Libraries & Frameworks

Backend

- Django
- Django REST Framework
- FastAPI

Frontend

- React
- React Router
- Axios
- Tailwind CSS
- Redux / Context API
- Chart.js / Recharts

Notifications

- Firebase Cloud Messaging (FCM)
- SendGrid

Testing

- pytest
- Django Test Client
- Jest
- React Testing Library

Dev & Deployment Tools

- VS Code
 - Git & GitHub
 - Docker & Docker Compose
 - GitHub Actions
 - Render / AWS / Vercel
 - Postman
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8. Performance Metrics

Financial Management Metrics

- Expense tracking accuracy
- Budget utilization efficiency
- Savings goal completion rate
- Income tracking accuracy

Analytics Metrics

- Category-wise spending insights
- Monthly trend analysis accuracy
- Report generation success rate
- Financial summary accuracy

System Performance Metrics

- API response time
 - Dashboard loading speed
 - Notification delivery success rate
 - Concurrent user handling capacity
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9. Example Quantitative Goals

Expense Management

Track and categorize daily expenses accurately.

Budget Planning

Monitor monthly budget utilization and prevent overspending.

Savings Goals

Help users achieve savings targets through progress tracking and reminders.

Financial Analytics

Provide actionable financial insights through dashboards and reports.

Platform Performance

Support multiple users and financial transactions while maintaining stable performance and responsiveness.